

CHAPTER 12: OPEN SPACE & ENVIRONMENTAL PROTECTION

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Section 12.1 Purpose and Applicability

The purpose of this chapter is for the protection of existing environmental resources including open space, streams, wetlands, watersheds, floodplains, soils, forest stands, specimen trees and other significant vegetation and wildlife. These elements are of economic value to the Town and make it a desirable place to live and visit.

The Open Space requirements of this section apply only to new developments with greater than 10 residential dwelling units, except for Open Space Subdivisions within R-S Districts, which are required to meet open space requirements regardless of the number of lots proposed. The C-B District and developments in which all lots are greater than one acre are exempt from this provision. Non-residential developments are encouraged to preserve open space wherever possible. All other requirements of this Chapter apply to all new developments and construction in all districts.

Section 12.2 Types of Open Space

All required open space shall be classified in accordance with this Section.

The following open space types may be either common area open space or public open space:

- A. **Playground**-Playgrounds are for active recreational use and provide sunny and shaded play equipment and play areas for children as well as open shelter with benches. Playgrounds may be part of other types of open space, such as parks, or may stand alone. Playgrounds may be fenced, lockable and lit if not part of a park.

Minimum Size: 3,000 square feet

Maximum Size: 20,000 square feet

- B. **Square**-Squares are areas for passive recreational use. Squares shall be bounded by streets on a minimum of three sides or 75 percent of their perimeter. Squares are encouraged to be entirely bounded by streets and/or lanes. Squares may be bounded by buildings on a maximum of 60 percent of their perimeter [maximum of two (2) sides] if through design, central gathering area for the area is formed. Squares shall be planted parallel to all streets and shall contain canopy trees along street frontages.

Minimum size: 2,000 square feet

Maximum size: 1 acre

- C. **Plaza**- A plaza is an open area adjacent to, or part of, a non-residential building or facility. Plazas function as gathering places and may incorporate a variety of non-permanent activities such as vendors and display stands. Plazas are generally paved in brick or other type of paver or crushed stone. Plazas shall be level, stepped, or gently sloping [less than five (5) percent grade]. Plazas may be left unplanted. If planted, the trees should form a frame around the plaza space. The following sizes are recommended but may be larger or smaller depending on the building or facility design.

Minimum size: 2,000 sq ft

Maximum size: 30,000 sq ft

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- D. **Park**-Parks may be designed for passive and/or active recreational use. Parks shall be bounded by streets on a minimum of 25 percent of their perimeter. Parks are encouraged to be entirely bounded by streets. Large parks should create a central open space which services an entire neighborhood or group of neighborhoods; or incorporates physical features which are an asset to the community (i.e. lake or river frontage, high ground, significant stands of trees). Undergrowth should be limited and landscaping shall be installed in a manner that promotes attractiveness and safety. Parks may be combined with greenways and greenbelts.

Minimum size: 1 acre

- E. **Green**- The green is an open space which is more natural. Like the square, it is small and surrounded by buildings. Unlike the square, it is informally planted and may have irregular topography. Greens are usually landscaped with trees at the edges and open lawns at the center. Greens should contain no structures other than benches, pavilions, and memorials; paths are optional.

Minimum size: 20,000 square feet

- F. **Greenway**-Greenways are large, irregular open spaces designed to incorporate natural settings such as creeks and significant stands of trees within and between neighborhoods. Greenways are typically more natural and may contain irregular topography. Greenways shall be used for certain active recreational uses including, at a minimum, trails for walking, jogging, and biking. Greenways shall connect points of interest in the community such as schools, parks, and other civic uses.

- G. **Greenbelt**-Greenbelts run along the perimeter of a neighborhood or Town, and serve to buffer a neighborhood from surrounding non-compatible uses such as a highway corridor or industrial district, or a Town from agricultural areas or adjacent towns. Greenbelts differ from the other types of open spaces in that they are left natural, and are not intended for recreational use.

- H. **Nature Preserve**-Open spaces designated as Nature Preserves shall be left largely undisturbed except for the optional clearing of underbrush and provision of a walking trail (mulch or other natural material only). Nature Preserve areas are encouraged to protect large stands of trees, wildlife, and natural water features.

Minimum size: 3 acres

- I. **Cluster Mailbox Court** – A type of open space whose primary purpose is to provide a publicly-accessible location for a development's cluster mailbox unit(s), where such units are required by the United States Postal Service (USPS).
- J. **Thoroughfare Open Space** – A type of open space intended to provide a buffer between Major and Minor Thoroughfares as designated in the Cabarrus-Rowan MPO's Metropolitan Transportation Plan and Major Subdivisions. This open space shall be provided along any section of the subdivision which is adjacent to said thoroughfare(s), and shall consist of a minimum 20' wide Type E Bufferyard meeting the composition requirements of Section 11.3.5.2.

Thoroughfare Open Space meeting the aforementioned standard shall not be required of subdivisions which utilize the Open Space Subdivision option (such subdivisions shall include open space along thoroughfares meeting the requirements of Section 7.3.4.4.C).

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Section 12.3 General Provisions for Open Space

- A. Land designated as open space on the approved development plan shall be maintained as open space and may not be separately sold, subdivided, or developed.
- B. Access from a public street shall be provided to all designated open space with a minimum 15-foot-wide access included in the open space area.
- C. Open space in the form of playgrounds, squares, plazas, parks, and greens shall be centrally located within a development.
- D. Open space may either be held in common ownership by a Homeowners Association or dedicated to the Town of China Grove, Rowan County, another appropriate public entity, or non-profit agency so long as the space is accepted by the entity and will be properly maintained by the entity.
- E. Common ownership of the open space requires that the Homeowners Association assume full responsibility for its maintenance. The restrictive covenants shall provide that, in the event the homeowners association fails to maintain the open space according to the standards of this Ordinance, the Town may, following reasonable notice, demand that deficiency of maintenance be corrected, or enter the open space to maintain it. The cost of such maintenance shall be charged to the homeowners' association.
- F. Standards for Cluster Mailboxes:
 - i. Cluster mailboxes shall be located within 25' of a parking area serving a development's clubhouse, community pool, or similar common amenity; or within 25' of the ROW of a street within the development.
 - ii. Two parking spaces shall be required for mailbox clusters of less than 50 mailboxes; three spaces shall be required for clusters of 50 mailboxes or more. These parking requirements shall not apply to cluster mailboxes located within 25' of parking areas for community amenities described above.
 - iii. Sidewalks or other all-weather surfaces shall connect cluster mailboxes to the community's sidewalk network.

Section 12.4 Minimum Open Space Dedication

Open space shall be dedicated in accordance with the below table. Percentages are based on total development area. Areas within floodplains, stream buffers, utility rights-of-way, and other areas that cannot be developed shall not be used in the total development area in order to calculate minimum open space dedication. However, these areas may be used to meet open space requirements if they fit into one of the open space categories listed in Section 12.2.

Section 12.5 Tree Preservation and Environmental Protection

12.5.1 Applicability

Significant forest stands, natural vegetation, specimen trees, severe natural topography, drainage features and water courses are encouraged to be preserved to the extent that is reasonable and practical while otherwise not reasonably prohibiting development. Forested and vegetated areas whose physical site conditions render them unsuitable for development should be set aside as conservation areas or as open

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space. Wooded sites should be developed with careful consideration of the natural characteristics of the site. When portions of forested stands must be developed, careful consideration should be given to preserving wooded perimeters or the most desirable natural features in order to retain the aesthetic or visual character of the site. Isolated pockets of existing trees or specimen trees should be protected as a valuable asset of the property. This section shall apply to all zoning districts regulated by this Ordinance.

Zoning District	Percent Open Space
R-P	n/a
R-S	10%; 20% for Open Space Subdivisions developed in accordance with Section 7.3.4
R-T	15%
R-M	20%
R-MH	15%
O-I	15%
N-C	20%
C-B	n/a
H-B	n/a
C-P	n/a
L-I	n/a
H-I	n/a

Determination of the need to preserve significant vegetation on the site is also a function of a site's buildable area or potential for development defined as follows:

- *Prime Buildable.* Land with little or no building restrictions that occur as a result of slope conditions or site topography. These areas are defined as slopes less than 10 percent and generally offer the least opportunity for the preservation of existing tree canopy, forest stands, or significant vegetation outside of satisfying open space dedication requirements.
- *Secondary Buildable.* Areas with slopes of 10 to 15 percent, site preparation techniques should minimize grading. Such areas require selective clearing and grading. These areas offer limited opportunity for the preservation of existing tree canopy, forest stands, or significant vegetation outside of satisfying open space dedication requirements.
- *Conserved.* Areas with slopes of 15 to 25 percent with severe slope restrictions. These areas offer optimal opportunity for the preservation of existing tree canopy, forest stands, or significant vegetation outside of satisfying open space dedication requirements.
- *Preserved.* Natural floodplain and floodways, wetland areas, existing tree canopy, forest stands, or significant vegetation on slopes exceeding 25 percent, and which present severe or prohibitive slope conditions for development shall be preserved.

Development proposals incorporating tree preservation shall include such measures as part of the required landscaping plan as described in Section 11.2.

12.5.2 Environmental Survey

Environmental Surveys are required for all developments that have greater than one acre of disturbed area or otherwise require a soil and erosion control permit. Identification of existing trees, understory vegetation, known endangered species, wetlands, streams and creeks, floodplains, and topographical features on a site prior to the advanced preparation of development plans enables the reasonable and

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practical planned preservation of existing vegetation while considering unique site conditions. This requirement provides the Town and the applicant the ability to evaluate the proposed development in order to preserve vegetation, to improve the appearance of the development proposed and to encourage the use of the existing forest and tree canopy, specimen trees, and significant vegetation to satisfy the requirements of this Chapter. An environmental survey is intended to identify forest stands, distinctive tree lines or forest edges, existing watercourses, and previously documented endangered species habitats. A zoning permit shall not be issued for developments of greater than one acre of disturbed area unless the Town first approves an Environmental Survey.

Environmental survey requirements are as follows:

- A. The area to survey is to be located within the proposed construction limits, building envelope, or area of the site to be disturbed and is to extend a minimum of 10 feet outside of this area.
- A. Denote the dripline of any existing forest stand, as measured between existing tree trunks six (6) inches or greater located at the edge of the stand.
- A. Show all other important natural features influencing site design such as the location of wetlands, rock outcroppings, site topography at two-foot contour intervals, slopes steeper than 15 percent, and perennial streams, natural drainageways, lakes, and other water bodies.
- B. Show any existing watercourses including wetlands, streams, creeks, ponds, and floodplains indicating both the flood fringe and the flood way.
- C. Denote the presence of any known endangered species indicated in any surveys completed by the State of North Carolina or other governmental agency.

12.5.3 Tree Removal Permits

- A. No person shall remove, destroy, cut, or severely prune any tree or shrub having its trunk in or upon any public property or on any street right-of-way or any tree or shrub on an approved site plan without first obtaining a permit from the Zoning Administrator. A permit shall also be issued for the removal of any tree on public or private property that is listed as a state or national champion by the North Carolina Forestry Service or the American Forestry Association; or if it provides habitat for endangered wildlife protected by federal law; or has been cited by the Town as being historically significant. Public and private utilities shall submit written specifications for pruning and trenching operations to the Town for approval.
- B. Exemptions from tree removal permits:
 - Single-family detached residential lots
 - Two-family attached (duplex) residential lots
 - The Town of China Grove and the NCDOT as long as tree preservation and protection requirements are included in the project plans.
 - Utility companies, except that written specifications for pruning and trenching shall be submitted to the Town for information purposes.
- C. The Town of China Grove may seek compensation from the property owner for the removal of hazardous vegetation. The Public Works Director is authorized to remove trees that are dead, diseased, irreparably damaged, or hazardous and are creating damage and/or are a threat to public safety. The Public Works Director shall be authorized to bill the owner for such services provided that the owner had been notified of the hazardous conditions and failed to act. The Public Works Director, upon

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determination that a tree on private property creates a threat to public safety, shall notify the property owner in writing to preserve or remove the hazardous tree. In the event the property owner does not respond to this notification, the Town of China Grove may seek civil remedy under the provisions of this Ordinance and may enter the property to remove the hazardous vegetation.

Section 12.6 Watershed Protection Requirements

As detailed in Section 7.16 Watershed Overlay District, applicable development activities in the Lake Fisher and Lake Corriher watershed consistent with the WS-IV Protected Area management rules as adopted by the North Carolina Department of Environment and Natural Resources, shall apply. These rules include provisions for roads and bridges; riparian buffers along perennial streams, rivers and impoundments; cluster development; how existing development and pre-existing lots are handled; density/impervious surface limitations; and Special Intensity Allocations. Any new subdivisions created within the watershed protection area which contain perennial streams, rivers and impoundments subject to riparian buffer requirements, shall include the required riparian buffer on the plat, as per NCGS 143-214.23A(f).